

VU KHANH CAO

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EDUCATION

Carleton University, Ottawa, ON

Sept 2023 – April 2028

Bachelor of Computer Science (Honours) – Artificial Intelligence and Machine Learning

GPA: 3.75 / 4.00

EXPERIENCE

Software Developer & DevOps Developer

May 2026 – Present

Bank of Canada

Ottawa, ON

- Led development and QA efforts to modernize the Treasury Management Systems (TMS) using **Java Spring Boot** and **TypeScript**, supporting over **\$1 trillion** in transactions annually.
- Designed and developed an **agentic code review suite** integrated into Azure CI/CD pipelines, improving compliance with **strict security and quality standards** across **mission-critical systems** by 60%.
- Pioneered the migration of legacy applications to **OpenShift**-based cloud infrastructure, improving **scalability, reliability, and deployment efficiency** for systems supporting **2,000+ employees**.

Software Developer

May 2025 – Dec 2025

Ovenns Analytics Inc.

Ottawa, ON

- Drove end-to-end development of web platforms adopted by **10+ businesses**, contributing across the full product lifecycle from **market research and UX design** to implementation, deployment, and maintenance.
- Architected and developed **secure, scalable backend services** using **Node.js, Express, and PostgreSQL**, supporting **100+ API endpoints**, multi-tenant access, and reliable data workflows.

PROJECTS

GURT – [GitHub](#)

Jan 2026

- Engineered a **real-time robotic control stack** spanning QNX 8, Raspberry Pi 4, Arduino motor control, and WebSockets, enabling live video streaming and **sub-100ms** remote command execution.
- Developed a custom **QNX 8 camera driver** that increased video throughput from **~3 FPS to a stable 30 FPS (10x)**, contributing to a **Best Hardware Hack** win at uOttaHack 8.

JARVIS – [GitHub](#)

Jan 2025

- Built a low-cost **physical-to-digital reconstruction pipeline** using QNX, Raspberry Pi, and Gaussian Splatting to transform multi-camera imagery into web-ready 3D scenes, reducing capture hardware cost from **\$400 to \$20**.
- Containerized the reconstruction pipeline with **Docker** and integrated **AWS S3 and GPU processing** to automate image ingestion, model training, and web delivery of generated 3D scenes.

TRACY – [GitHub](#)

Feb 2024

- Developed a **monocular computer vision pipeline** using TrackNetV2 and convolutional RNNs to reconstruct player and ball trajectories from a **single camera**, eliminating the need for specialized hardware.
- Built gameplay analytics for **serve speed, ball bounces, player heatmaps, and in/out detection** using Matplotlib and FFmpeg, contributing to a **3rd-place finish at QHacks 2024**.

SKILLS

Languages

Python, Java, JavaScript, TypeScript, C/C++, SQL

Libraries & Frameworks

React, Angular, Node.js, Express, Spring Boot, TensorFlow, OpenCV

Platforms & Technologies

Azure, AWS, Docker, Kubernetes, ArgoCD, OpenShift, REST APIs, Git